Probabilistic Hierarchical Retention Time Prediction for LC–MS

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1 Introduction

This document describes a **probabilistic, multi-species, multi-compound hierarchical** model to predict LC–MS retention times for each (species, compound, run) combination, together with a calibrated uncertainty. A simple baseline is to fit an independent supervised model (for example, a lasso regression) for every species (or species group)–compound pair using run-level features. In practice this "one model per pair" strategy fragments the data, fails to share information across related species and compounds, behaves poorly for rare or newly added combinations, and tends to overfit run covariates instead of capturing global patterns of chromatographic drift.

We instead use a single hierarchical, multi-task regression model. Different species and compounds are treated as related tasks that share information through global priors: strongly supported effects can deviate from the average, while poorly observed groups are shrunk gently toward a common pattern. This improves robustness in small-data regimes and yields sensible behavior for rare or unseen combinations.

Two ingredients drive this sharing in the model family. First, each compound can be represented by a *descriptor*: a fixed-length numeric vector summarizing its chemistry (for example, a PCA-compressed neural embedding or a fingerprint-derived feature vector based on its SMILES string). Working in this shared descriptor space lets the model generalize across chemically similar compounds and provide calibrated priors for sparsely observed or new compounds. In the production configuration used for the cap-based library runs and the primary figures in this document, we encode chemistry directly via descriptors (for example, ChemBERTa PCA-20), and use discrete chemical classes only to share run-covariate patterns.

Second, we include *run covariates*: per-run features that summarize LC–MS operating conditions. In the current system these covariates are derived from internal-standard channels; endogenous standards and additional run-level signals can be folded in the same way as they become available. These covariates allow the model to explain systematic run-to-run shifts without training a separate model for every run.

Taken together with the experiments in later sections, our current evidence suggests that **a single hierarchical RT model, with chemistry encoded via descriptors and classes used only to share run-sensitivity patterns, is the right direction for generalisation across chemistry, rare compounds, and shifted runs**. The hierarchical structure and descriptor-based chemistry encoding improve behaviour for rare species and sparse chemistry, perform better than per-pair

lasso in diverse species groups such as cells, and show greater robustness to the kinds of run-level shifts we expect when deploying kits and moving from Thermo-trained libraries to new instruments such as Agilent. These capabilities are central to the RT work planned for 2026.

2 Hierarchical Retention Time Model

2.1 Model specification

At the observation level we work with simple (s, c, r) indices:

- s = species (or species/matrix group),
- c = compound,
- r = LC–MS run,
- y_{scr} = observed RT for species s, compound c, run r,
- $\boldsymbol{x}_r^\star \in \mathbb{R}^P$ = standardized run-covariate vector for run r (for example, internal-standard channels).

In these indices, the production configuration of the hierarchical RT model can be written compactly as

$$y_{scr} = \mu_0 + \alpha_s + \beta_c + \boldsymbol{w}_c^\top \boldsymbol{x}_r^\star + \epsilon_{scr}, \qquad \epsilon_{scr} \sim \mathcal{N}\big(0, \sigma^2_{y,k(s)}\big), \tag{1}$$

$$\alpha_s \sim \mathcal{N}\big(\mu^{(\text{cluster})}_{k(s)}, \sigma^2_{\text{species}}\big), \qquad \mu^{(\text{cluster})}_k \sim \mathcal{N}\big(0, \sigma^2_{\text{cluster}}\big), \tag{2}$$

$$\beta_c = \boldsymbol{z}_c^{\star\top} \boldsymbol{\theta}_\beta + \delta_c, \qquad \delta_c \sim \mathcal{N}\big(0, \sigma^2_{\text{compound}}\big), \tag{3}$$

where $k(s)$ maps each species to a species cluster, $\boldsymbol{z}_c^\star \in \mathbb{R}^d$ is the standardized descriptor vector (for example, ChemBERTa PCA-20) for compound c, and $\{\boldsymbol{w}_c\}$ are compound-specific covariate weights. In words: μ_0 is a global intercept (grand mean RT across species, compounds, and runs), α_s is a species-level adjustment, β_c is a descriptor-informed compound baseline, and $\boldsymbol{w}_c^\top \boldsymbol{x}_r^\star$ captures how compound c responds to run-to-run shifts encoded by $\boldsymbol{x}_r^\star$.

The prior choices reflect weak information on a standardized scale and act as a form of regularization. They keep group-level effects from becoming unrealistically large in poorly observed settings, while still allowing the data to dominate when there is ample evidence. In the current implementation we use HalfNormal priors on the key scale parameters and Normal priors on fixed effects; these encourage shrinkage toward zero without ruling out larger values when strongly supported by the data.

Variance components:

$$\sigma_{\text{cluster}} \sim \text{HalfNormal}(0.5), \tag{4}$$

$$\sigma_{\text{species}} \sim \text{HalfNormal}(0.5), \tag{5}$$

$$\sigma_{\text{compound}} \sim \text{HalfNormal}(0.5). \tag{6}$$

The observation-noise scale is *cluster-specific* in the production model: for each species cluster k we use

$$\log \sigma_{y,k} \sim \mathcal{N}(\log 0.02,\, 0.4^2), \qquad \sigma_{y,k} = \exp(\log \sigma_{y,k}), \tag{7}$$

which corresponds to a log-normal prior centered near a typical within-cluster noise scale (minutes) with moderate spread. For convenience in summaries we also define a single scalar σ_y as a root-mean-square aggregation over clusters.

For the global intercept and descriptor weights we use

$$\mu_0 \sim \mathcal{N}(\bar{t}, 5.0), \qquad \tau_\beta \sim \text{HalfNormal}(0.6), \qquad \boldsymbol{\theta}_\beta \sim \mathcal{N}\big(\mathbf{0}, \tau_\beta^2\, \boldsymbol{I}_d\big), \tag{8}$$

where $\bar{t}$ is the empirical mean RT in the training data. This places a weakly informative prior on the global intercept and on the descriptor-to-RT mapping.

For run-covariate weights, the implementation works with a compound-by-covariate matrix γ rather than an explicit $\boldsymbol{w}_c$. When covariates are present, we use

$$\sigma_\gamma \sim \text{HalfNormal}(0.5), \qquad \gamma_{\text{raw}}[c,p] \sim \mathcal{N}(0,1), \tag{9}$$

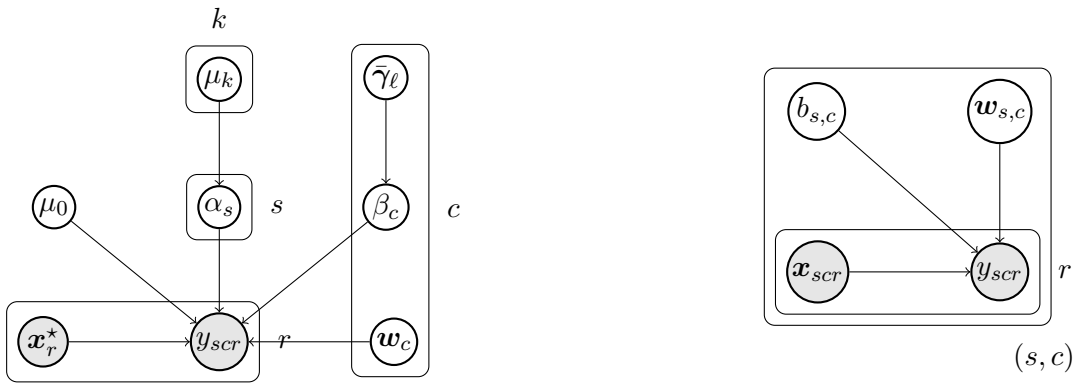
and set $\gamma[c,p]$ to a shrunk version of $\gamma_{\text{raw}}[c,p]$ around a class-specific mean:

$$\gamma_{\text{class}}[\ell,p] \sim \mathcal{N}(0, \sigma^2_{\gamma,\text{class}}), \qquad \gamma[c,p] = \gamma_{\text{class}}\big(g(c),p\big) + \sigma_\gamma\, \gamma_{\text{raw}}[c,p]. \tag{10}$$

This encourages compounds in the same chemistry group to share covariate patterns, while the half-normal priors on the relevant scales encourage small covariate effects unless the data provide strong evidence for larger shifts.

The implementation uses non-centered parameterisations and sum-to-zero recentering for the effects $\{\alpha_s\}$ and $\{\beta_c\}$ so that μ_0 remains the grand mean; these reparameterisations do not change the implied priors in (2)–(3). In the production configuration, the covariate weights $\boldsymbol{w}_c$ follow the class-pooled hierarchy above, so that compounds within the same chemical class share information about their run sensitivities. Because the covariates are standardized (mean zero, unit variance per component), the scale of the weights is interpretable and the model is numerically well behaved.

2.2 Lasso baseline and comparison



(a) Hierarchical RT model

(b) Per-(group, compound) lasso

Figure 1: Plate diagrams for the hierarchical RT model (left) and the per-(group, compound) lasso baseline (right). Shaded circles denote observed quantities (RT measurements y_{scr} and run covariates $\boldsymbol{x}_r^\star$); unshaded circles denote latent variables and parameters. Indices: s indexes species (or species/matrix groups), c indexes compounds, r indexes runs, k indexes species clusters, and ℓ indexes chemical classes. In the hierarchical panel, μ_0 is a global intercept, μ_k are cluster means, α_s are species effects, $\bar{\gamma}_\ell$ are class-level mean run-covariate effects, β_c are descriptor-informed compound baselines, and $\boldsymbol{w}_c$ are compound-specific covariate weights combining with run covariates $\boldsymbol{x}_r^\star$ to generate each observation y_{scr}. Plates labelled k, s, c, r indicate replication over clusters, species, compounds, and runs. In the lasso panel, each (s, c) has its own intercept $b_{s,c}$ and covariate weights $\boldsymbol{w}_{s,c}$ fitted independently, with the plate labelled (s, c) denoting one separate regression per species–compound pair. For clarity, not all hyperparameters (such as the scale parameters $\sigma_\cdot$) or derived quantities are drawn; their roles are described in the surrounding text.

For comparison, the current lasso baseline used elsewhere in the pipeline fits a separate penalised regression for each species (or matrix group) and compound. Fix a species/matrix group s and compound c, and let r index runs with feature vectors $\boldsymbol{x}_r^\star$ constructed from internal-standard and, when available, endogenous-standard channels (columns such as "IS*", "ES*", and related summaries). The lasso baseline fits an independent linear model

$$y_{scr} = b_{s,c} + \left(\boldsymbol{x}_r^\star\right)^\top \boldsymbol{w}_{s,c} + \epsilon_{scr}, \qquad \epsilon_{scr} \sim \mathcal{N}(0, \sigma_{s,c}^2), \tag{11}$$

with parameters found by solving the penalised least-squares problem

$$(\hat{b}_{s,c}, \hat{\boldsymbol{w}}_{s,c}) = \arg\min_{b,\boldsymbol{w}} \sum_r \left(y_{scr} - b - \left(\boldsymbol{x}_r^\star\right)^\top \boldsymbol{w}\right)^2 + \lambda \|\boldsymbol{w}\|_1. \tag{12}$$

Each (s, c) pair is treated in isolation: there is no sharing of information across groups, across compounds, or across related covariate patterns beyond the choice of penalty λ.

The hierarchical RT model can be viewed as a structured generalisation of this baseline. For the same species/matrix group s, compound c, and run r, it uses the mean structure in (1) together with the hierarchies in (2)–(3). Informally, the lasso parameters $b_{s,c}$ and $\boldsymbol{w}_{s,c}$ play a similar role to the collapsed hierarchical intercept $\mu_0 + \alpha_s + \beta_c$ and the compound-specific weights $\boldsymbol{w}_c$, but

without any sharing across groups or compounds. Equivalently, we can view the hierarchical model as inducing effective per-pair coefficients

$$b_{s,c}^{(\text{hier})} = \mu_0 + \alpha_s + \beta_c, \qquad \boldsymbol{w}_{s,c}^{(\text{hier})} = \boldsymbol{w}_c, \tag{13}$$

so the hierarchical mean can be written in the same form as the lasso, but with $b_{s,c}^{(\text{hier})}$ and $\boldsymbol{w}_{s,c}^{(\text{hier})}$ tied across (s, c) via shared priors on μ_0, α_s, β_c, and $\boldsymbol{w}_c$. This shared structure lets rare species–compound pairs borrow strength from well-sampled ones, while still behaving like per-pair regressions when data are abundant.

2.3 Molecular descriptors (20-dim SMILES embedding)

In the descriptor-augmented variant of the model we use per-compound molecular descriptors $\boldsymbol{z}_c \in \mathbb{R}^d$ with a fixed dimensionality $d = 20$ to enable sharing across chemically similar compounds and to provide calibrated priors for rarely observed or unseen compounds. Descriptors are computed offline (for example, from a pretrained SMILES encoder or from fingerprint features) and then passed into the model as a numeric matrix.

Before fitting we standardise each descriptor dimension across compounds to mean 0 and variance 1. This internal standardisation keeps the prior on $\boldsymbol{\theta}_\beta$ well-scaled and ensures that no single descriptor coordinate dominates purely because of its units. With centered descriptors, μ_0 remains interpretable as the grand mean RT. When descriptors are missing, we impute $\boldsymbol{z}_c$ to the zero vector in this standardised space; in that case the model relies more heavily on the residual scale σ_{compound} to represent uncertainty.

For typical library sizes, $d = 20$ strikes a balance between expressiveness and sampling stability. Larger descriptor dimensions increase cost and may require stronger regularization without improving generalization, so in practice one can choose d empirically and prefer the smallest value that matches held-out performance while maintaining good sampler diagnostics.

Why this helps rare compounds. When descriptors are enabled, a rarely observed compound c has its posterior baseline β_c shrink toward the descriptor-predicted mean $\boldsymbol{z}_c^\top \boldsymbol{\theta}_\beta$ rather than a broad class average. Uncertainty propagates both the mapping uncertainty in $\boldsymbol{\theta}_\beta$ and the residual scale σ_{compound}, yielding calibrated predictions.

3 Inference, Uncertainty, and Diagnostics

In practice we fit this model with the No-U-Turn Sampler (NUTS), using multiple chains and standard adaptations of step size and tree depth. When a JAX/NumPyro backend is available it can be used for faster gradient evaluations; otherwise the default PyMC NUTS implementation is used. Typical runs involve a few hundred to a few thousand posterior draws after warmup, with the exact configuration controlled by the sampling arguments (`draws`, `tune`, `target_accept`, and `max_treedepth`).

Given posterior draws, we summarise RT predictions for each species–compound pair (s, c) by reconstructing the linear predictor for a grid of runs and then computing the predictive mean and

variance. Writing t_{sc} for the model's implied RT for species s and compound c, the implementation follows

$$\hat{t}_{sc} = \mathbb{E}[t_{sc}], \qquad \hat{\sigma}^2_{sc} = \mathrm{Var}[t_{sc}] + \mathbb{E}[\sigma^2_{y,k(s)}], \tag{14}$$

where the expectation and variance are taken over posterior draws. The first term captures parameter uncertainty in the intercept, species, compounds, and covariate weights, while the second term adds back the average observation-noise variance $\sigma^2_{y,k(s)}$ to form a full predictive variance. At prediction time, helper routines reconstruct the linear predictor for requested (species, compound, run) tuples using posterior draws of $(\mu_0, \alpha_s, \beta_c, \boldsymbol{w}_c, \sigma_{y,k(s)})$ and return both mean predictions and associated standard deviations $\hat{\sigma}_{sc}$.

Standard MCMC diagnostics (e.g. absence of divergences, $\hat{R} \leq 1.01$, $\mathrm{ESS}_{\mathrm{bulk}} \geq 200$) are enforced when assessing model fits. The non-centered parameterisation and sum-to-zero recentering described in Section 4.3 are chosen precisely to make these diagnostics easier to satisfy in realistic LC–MS datasets.

Posterior predictive checks (PPC) are used to evaluate how well the model's predictive distribution matches observed RTs. In the reference implementation we draw samples from the posterior predictive distribution of y_{obs}, compute per-row predictive means, standard deviations, and central intervals (e.g. 95%), and then report summary metrics such as RMSE, MAE, empirical coverage of the 95% interval, and a small coverage curve at several nominal levels (0.5, 0.8, 0.9, 0.95). Simple residual diagnostics (observed minus predictive mean) are also retained for downstream plots and checks by species or chemistry group.

4 Implementation details: data, training, and evaluation

This section briefly describes how we construct the training datasets, fit the model, and evaluate it against the existing lasso baseline. The aim is to clarify the experimental setup rather than to spell out every engineering detail.

4.1 Data preparation and capping

For each library we start from a large merged RT dataset containing all labeled peaks, with metadata for species, compound, matrix type, run identifiers, and internal standards. We first restrict this dataset to rows with valid chemical classes and a valid species mapping so that the downstream models work on a coherent set of compounds and species.

We then apply a cap on the number of observations per species–compound–matrix combination. If the cap is N, at most N rows are retained for each such combination, with rows selected at random using a fixed seed. This keeps the set of species–compound pairs fixed while controlling how many replicate RT measurements each pair contributes. Larger caps (for example, 20 or 50) therefore provide more replicates per pair without changing which pairs are present.

After capping, we aggregate the internal-standard panels to form run-level covariates and assemble an RT training table with one row per labeled peak. This table is the input to the hierarchical model described in the earlier sections.

4.2 Model training on capped datasets

For each library and cap level we train a separate hierarchical RT model. The training routine reads the capped RT table, constructs integer index spaces for species and compounds, defines run indices from the run identifiers, and builds the standardized run covariate vectors $\boldsymbol{x}_r^\star$ from the internal-standard measurements.

The hierarchical model is then fitted with NUTS using moderate chain and draw counts chosen to balance runtime and diagnostics. Because the cap is enforced before training, the model always sees the same structure (species, compounds, runs); only the number of replicate observations per species–compound pair changes with the cap.

4.3 Implementation details and non-centered parameterization

This subsection is about sampler behaviour rather than modelling ideas; it is safe to skip on a first read. Conceptually, the hierarchical model is fully described by the centered priors in Section 2.2, for example

$$\alpha_s \sim \mathcal{N}\big(\mu_{k(s)}, \sigma^2_{\text{species}}\big), \qquad \beta_c \mid \mu_{g(c)} \sim \mathcal{N}\big(\mu_{g(c)},\ \sigma^2_{\text{compound}}\big). \tag{15}$$

Sampling these *centered* group effects directly is intuitive but can lead to "funnel"-shaped posteriors when the group-level scales (such as σ_{species} or σ_{compound}) are small. In those regions, the sampler must explore very narrow, highly correlated directions, which can cause slow mixing and divergent transitions for gradient-based methods like NUTS.

To avoid this we reparameterise the same model in a *non-centered* form. Instead of drawing α_s and β_c directly, we draw standardised latent variables and scale them by the corresponding standard deviations. For the cluster and species effects we write

$$\mu_k = \tilde{\mu}_k\, \sigma_{\text{cluster}}, \quad \tilde{\mu}_k \sim \mathcal{N}(0,1), \qquad \alpha_s = \mu_{k(s)} + \tilde{\alpha}_s\, \sigma_{\text{species}}, \quad \tilde{\alpha}_s \sim \mathcal{N}(0,1), \tag{16}$$

and for the compound effects we use

$$\tilde{\beta}_c \sim \mathcal{N}(0,1), \qquad \beta_c = \mu_{g(c)} + \tilde{\beta}_c\, \sigma_{\text{compound}}. \tag{17}$$

Mathematically this leaves the joint prior over (α_s, β_c) unchanged, but it puts most of the curvature into the simple standard-normal variables $(\tilde{\mu}_k, \tilde{\alpha}_s, \tilde{\beta}_c)$ and leaves the scale parameters $(\sigma_{\text{cluster}}, \sigma_{\text{species}}, \sigma_{\text{compound}})$ more weakly correlated. In practice this geometry is much easier for NUTS to explore and leads to fewer divergences.

As an additional implementation detail, we recenter the realised species effects so that they have mean zero across species. This "sum-to-zero" adjustment keeps μ_0 interpretable as the global RT level and helps separate overall shifts from species-specific deviations, but it does not change the predictive distribution.

4.4 Model implementation in `hierarchical.py`

Equations (1) and (13) describe the hierarchical RT model at the level of (s, c, r) indices and effective per-pair coefficients. The production implementation in `src/compassign/rt/hierarchical.py`

follows this specification closely, using PyMC random variables for each component of the hierarchy and a non-centered parameterisation for group effects.

Observation model ((1)). For convenience we restate the observation equation

$$y_{scr} = \mu_0 + \alpha_s + \beta_c + \boldsymbol{w}_c^\top \boldsymbol{x}_r^\star + \epsilon_{scr}, \qquad \epsilon_{scr} \sim \mathcal{N}\big(0, \sigma^2_{y,k(s)}\big),$$

which the code implements directly. Inside `HierarchicalRTModel.build_model`, the linear predictor corresponding to y_{scr} is assembled as

```
rt_mean = (
    mu0
    + species_eff[species_idx]
    + compound_eff[compound_idx]
    + cov_term
    + sc_term
)
pm.Normal(
    "y_obs",
    mu=rt_mean,
    sigma=sigma_y_group[cluster_idx],
    observed=obs_df["rt"].values,
)
```

Here `species_idx`, `compound_idx`, and `run_idx` are the integer indices for (s, c, r), `mu0` implements μ_0, `species_eff` implements α_s, `compound_eff` implements β_c, and `cov_term` implements $\boldsymbol{w}_c^\top \boldsymbol{x}_r^\star$. The term `sc_term` is an optional species–compound interaction intercept (enabled via `--species-compound-intercept`). The vector `cluster_idx` is derived from `species_idx` via the species-to-cluster mapping and selects the appropriate group noise scale $\sigma_{y,k(s)}$. The final line defines $y_{scr} \sim \mathcal{N}(\texttt{rt_mean}, \sigma^2_{y,k(s)})$, matching the likelihood in (1).

Group hierarchies. The species and cluster hierarchy in (2) is implemented via non-centered effects:

```
sigma_cluster = pm.HalfNormal("sigma_cluster", sigma=0.5)
sigma_species = pm.HalfNormal("sigma_species", sigma=0.5)
cluster_raw = pm.Normal("cluster_raw", 0.0, 1.0, shape=self.n_clusters)
cluster_eff = cluster_raw * sigma_cluster
cluster_eff = cluster_eff - pm.math.mean(cluster_eff)

species_raw = pm.Normal("species_raw", 0.0, 1.0, shape=self.n_species)
species_base = cluster_eff[self.species_cluster] + species_raw * sigma_species
species_eff = species_base - pm.math.mean(species_base)
```

Drawing `cluster_raw` and `species_raw` from standard normals and scaling by `sigma_cluster` and `sigma_species` gives the same Gaussian priors as in (2), while subtracting the mean enforces the

sum-to-zero recentering described earlier. The compound baseline in equation (3) is constructed as a descriptor term plus a compound residual, `desc_term + delta_c`; when `include_class_hierarchy=True`, an additional class-pooled component `class_eff[self.compound_class]` is added before recentering to produce `compound_eff`.

Run-covariate effects $\boldsymbol{w}_c$ are represented by the `gamma` tensor and the standardised covariate matrix. For each observation the code computes `cov_term` as

$$\texttt{cov_term}_i = \sum_p x^\star_{r(i),p}\, \gamma_{c(i),p},$$

which is the dot product $\boldsymbol{w}_c^\top \boldsymbol{x}_r^\star$ in (1) implemented as an elementwise product and sum over covariate dimensions.

Effective per-pair coefficients ((13)). The effective per-pair coefficients are given by

$$b_{s,c}^{(\text{hier})} = \mu_0 + \alpha_s + \beta_c, \qquad \boldsymbol{w}_{s,c}^{(\text{hier})} = \boldsymbol{w}_c,$$

and `HierarchicalRTModel.predict_new` reconstructs these per-(species, compound) coefficients from posterior draws:

```
mu0 = posterior["mu0"].values.flatten()
species_eff = posterior["species_eff"].values.reshape(-1, self.n_species)
compound_eff = posterior["compound_eff"].values.reshape(-1, self.n_compounds)
...
for i in range(n_samples_used):
    pred = (
        mu0[i]
        + species_eff[i, species_idx]
        + compound_eff[i, compound_idx]
    )
    if has_covariates:
        pred = pred + np.sum(cov_std * gamma[i][compound_idx], axis=1)
```

For each posterior draw i, the term in parentheses plays the role of $b_{s,c}^{(\text{hier})} = \mu_0 + \alpha_s + \beta_c$, and the row `gamma[i][compound_idx]` provides the per-compound covariate weights $\boldsymbol{w}_c$ in (13). The resulting predictions therefore have exactly the hierarchical structure described in Section 2, but expressed as a collection of coupled per-pair regressions in code.

5 Results

5.1 Test set performance

To assess out-of-sample performance we evaluate the hierarchical model on separate real-test datasets that mirror the production schema but are held out from training. For each library we reload the trained model and its posterior draws, align the real-test species and compounds

to the training index space, and compute predictive means and uncertainties for all overlapping (species, compound, run) combinations.

In parallel we evaluate the existing per-pair lasso baseline on the same real-test data. Each lasso model is a separate regression for a particular chemistry supercategory and compound, with its own set of covariates and a prediction window. During evaluation we only obtain lasso predictions for those compounds that have a corresponding fitted model, so the lasso coverage can be substantially below one in some species groups.

Figure 2 summarises the comparison for one representative setting: library 208, where the hierarchical model is trained on a cap-50 dataset and evaluated on its real-test set alongside the lasso baseline. The top panel reports RMSE in minutes by species group, and the bottom panel shows the fraction of test-set compounds in each group for which each model produces a prediction. The hierarchical model achieves lower RMSE than the lasso in most species groups, with particularly clear gains for heterogeneous matrices such as fecal samples and human cells, while the lasso is only marginally better in human blood and nearly tied in urine. Across all groups the hierarchical model also maintains full compound coverage, whereas the lasso typically leaves 10–25% of compounds in each group without any model.

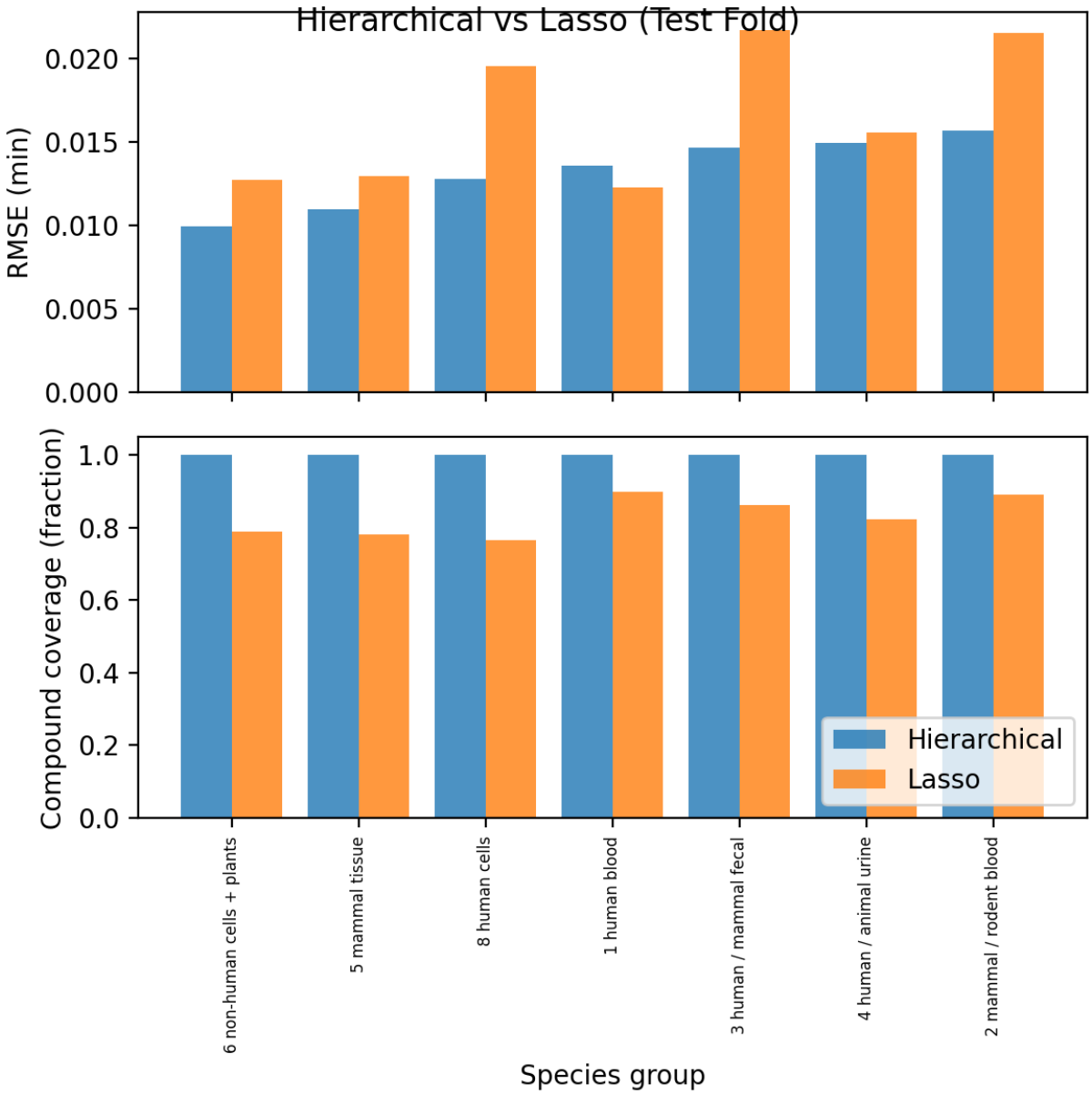


Figure 2: Hierarchical RT model (blue) versus per-pair lasso baseline (orange) on the lib208 real-test dataset when the hierarchical model is trained on a cap-50 RT dataset. The upper panel shows RMSE by species group; the lower panel shows the fraction of test-set compounds in each species group for which the model provides a prediction (compound coverage).

To give a sense of data usage, Table 1 reports how many capped training rows are used at different cap levels for libraries 208 and 209, and what fraction of the full merged dataset they represent. The cap-50 configuration underlying Figure 2 uses only about 14% of the available labeled peaks in lib208 (and about 2% in lib209) while keeping the full set of species–compound pairs in each library.

Library	Cap	Training rows	Species–compound pairs	Rows as % of full
208	5	80714	11054	1.58
208	10	160782	11054	3.15
208	20	315877	11054	6.20
208	**50**	**714672**	**11054**	**14.02**
208	100	1185761	11054	23.26
208	200	1792881	11054	35.17
208	500	2764070	11054	54.22
208	1000	3331202	11054	65.35
209	5	258772	51364	0.26
209	10	514122	51364	0.51
209	20	998902	51364	0.99
209	**50**	**2187265**	**51364**	**2.17**
209	100	3452547	51364	3.43
209	200	5202910	51364	5.16
209	500	8750350	51364	8.68
209	1000	12534514	51364	12.44

Table 1: Training set size for libraries 208 and 209 at different cap levels, taken directly from the cap summary. Increasing the cap adds more replicate RT measurements per species–compound pair without changing the set of pairs in each library.

Table 2 shows RT RMSE and compound coverage by species group for the same lib208 cap-50 experiment. On average across groups the hierarchical model reduces RMSE from about 0.0166 min (lasso) to 0.0132 min, with the largest relative gains in challenging matrices such as fecal samples and human cells. At the same time the hierarchical model maintains essentially full compound coverage in all groups, whereas the lasso baseline covers only around 76–90% of compounds per group.

Species group	RMSE (hier)	RMSE (lasso)	Coverage (hier)	Coverage (lasso)
1 human blood	0.0136	**0.0123**	1.00	0.90
2 mammal / rodent blood	**0.0157**	0.0215	1.00	0.89
3 human / mammal fecal	**0.0146**	0.0217	1.00	0.86
4 human / animal urine	**0.0149**	0.0156	1.00	0.82
5 mammal tissue	**0.0110**	0.0129	1.00	0.78
6 non-human cells + plants	**0.0099**	0.0127	1.00	0.79
8 human cells	**0.0128**	0.0196	1.00	0.76

Table 2: Per-species-group RMSE (minutes) and compound coverage for lib208, cap-50 training, evaluated on the real-test set. Coverage denotes the fraction of distinct compounds in each species group for which the model produces a prediction.

One notable exception is human blood, where the per-pair lasso retains a small RMSE advantage despite slightly lower coverage. This is consistent with the intuition that in very well sampled, relatively homogeneous settings a separate linear model for each species–compound pair is already close to optimal. In such regimes the hierarchical model trades a small amount of best-case RMSE for a simpler shared structure. Natural extensions to narrow this gap include adding a lightly reg-

ularised interaction term for each species–compound pair (a hierarchical intercept $\delta_{s,c}$ that shrinks to zero when data are sparse but allows per-pair flexibility when many observations are available) and increasing the training cap specifically for high-data groups such as human blood so that the hierarchy can better match the fine-grained structure that lasso exploits there.

5.2 Synthetic benchmarking of the RT hierarchy

Beyond real-data evaluations, we benchmark the hierarchical RT model and its descriptor-augmented variants on controlled synthetic datasets. These experiments are designed to probe specific questions that are difficult to answer cleanly in production data, such as how much descriptors help in sparse regimes, how well the model extrapolates to unseen compounds and chemistry classes, and how robust it is to covariate shift in run-level features.

5.2.1 Descriptor ablation on synthetic RT

The descriptor ablation experiment compares three arms of the hierarchical model on synthetic RT data: a full model with descriptors and run covariates; a model with descriptors but no covariates; and a model with covariates but no descriptors or class hierarchy. Synthetic compounds are stratified into anchor, rare, and unseen groups using the same descriptor space that feeds the model. Labels are allocated so that anchors are moderately well observed, rare compounds have only a handful of training labels, and unseen compounds are test-only.

For each replicate we fit all arms with matched sampler settings and report overall and group-wise error metrics, together with MAE versus nearest-anchor distance. This ablation isolates the contribution of descriptors over and above run covariates: in our runs, **both descriptor-enabled arms achieve similar overall MAE and clearly outperform the chemistry-free arm, with roughly 10–15% MAE reduction for rare compounds relative to the covariates-only arm**. For anchor compounds all three arms are virtually indistinguishable. In this particular synthetic setup, adding covariates on top of descriptors provides little extra gain, suggesting that chemistry is doing most of the work and that run covariates mainly help align well-supported compounds.

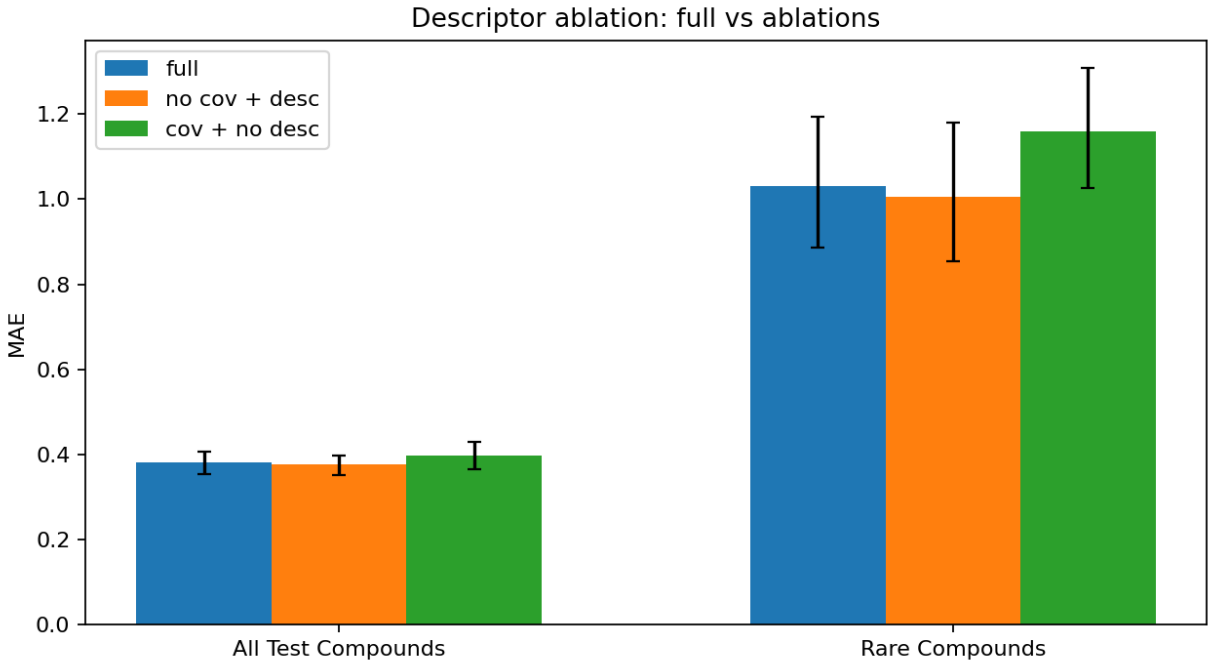


Figure 3: Descriptor ablation on synthetic RT. Mean MAE (with 95% bootstrap intervals) for three hierarchical arms—full (covariates + descriptors), descriptors without covariates, and covariates without descriptors—shown overall and for rare compounds. The descriptor-enabled arm achieves the lowest MAE for rare and unseen compounds while matching the chemistry-free arm when support is high.

5.2.2 Generalisation to unseen compounds and classes

A second set of experiments probes how descriptor-informed hierarchies generalise across chemistry. We consider two synthetic scenarios. In a cold-start setting, a subset of compounds is held out entirely from training and only evaluated at test time, while anchors and rare compounds are used for fitting. This mirrors the practical case where a new compound is added to a library: no labeled RTs for that compound are available when the model is trained, but we still need a reasonable RT prior based on descriptors and run covariates.

In a mix-shift setting, entire chemistry classes are held out from training and only evaluated at test time, so that the model must extrapolate across descriptor space rather than across runs. This corresponds to situations where the chemistry mix in a study differs substantially from the mix in the training library—for example, when a platform trained on mostly lipids is applied to a study enriched for steroids or phospholipids. The question is whether the descriptor hierarchy can transfer what it learns from well-sampled classes to structurally related but unobserved classes.

In both scenarios we compare the hierarchical RT model with and without descriptors, using the same underlying synthetic generator and training splits. We evaluate MAE, RMSE, bias, and tail error for unseen compounds and held-out classes versus their seen counterparts. **Across replicates the descriptor-enabled hierarchy reduces MAE for unseen compounds by roughly 30–40% compared with the no-descriptor hierarchy, with similar gains for held-out classes**

while leaving error on seen chemistry essentially unchanged or slightly improved. These synthetic results support the view that the descriptor prior is genuinely enabling extrapolation across chemistry rather than merely smoothing within well-sampled classes.

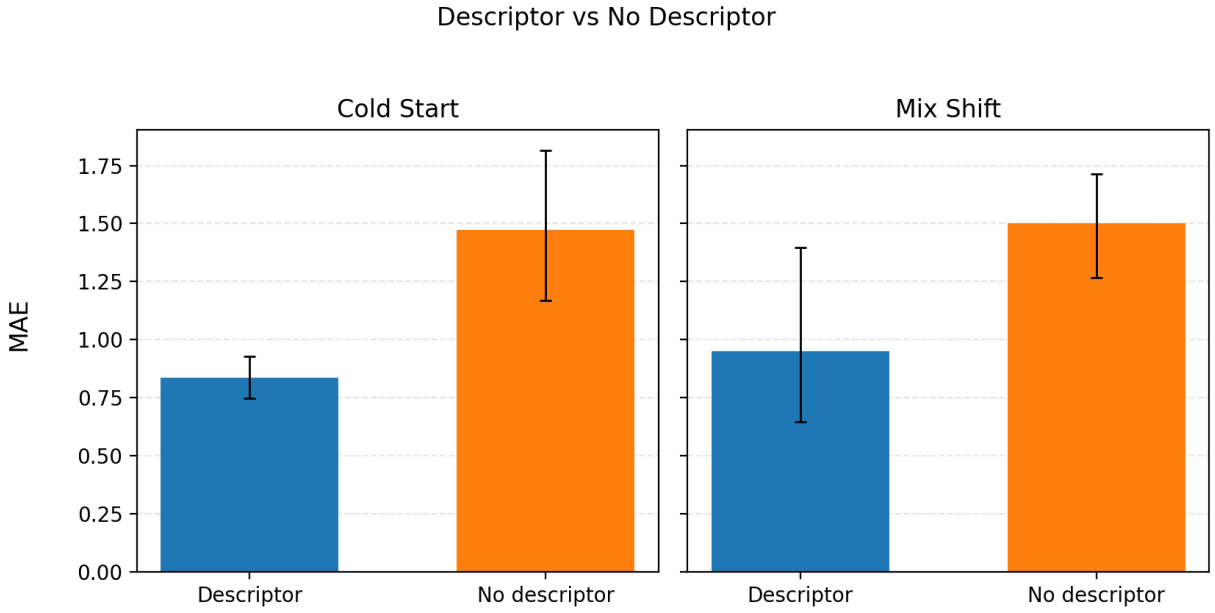


Figure 4: Descriptor generalisation on synthetic RT. Mean MAE for the hierarchical model with and without descriptors under two scenarios: unseen compounds (cold start) and held-out chemistry classes (mix shift). Descriptors substantially reduce MAE for truly unseen chemistry while leaving performance on seen compounds and classes unchanged or slightly improved.

5.2.3 Descriptor robustness under run-level drift

To study robustness under run-level covariate drift we generate synthetic datasets with internal-standard covariates and descriptors, fix a set of in-support runs for training, and evaluate several methods on held-out runs. We compare a hierarchical model with descriptors, a chemistry-free hierarchical model, and two lasso baselines that operate at the species–compound and cluster–compound levels. All methods are evaluated twice on the same held-out rows: once using the original run covariates and once after perturbing the covariates along an internal-standard-weighted drift direction.

The resulting MAE and RMSE curves show how each method degrades as drift is applied. **In our synthetic runs the descriptor-enabled hierarchy remains competitive with the best lasso baseline under observed covariates and becomes clearly superior once drift is applied: its MAE increases only slightly, whereas both the chemistry-free hierarchy and the baselines show noticeably larger degradation**. This suggests that descriptor-informed pooling helps stabilise predictions for runs that move modestly out of the training covariate support, although it does not remove the impact of large drifts.

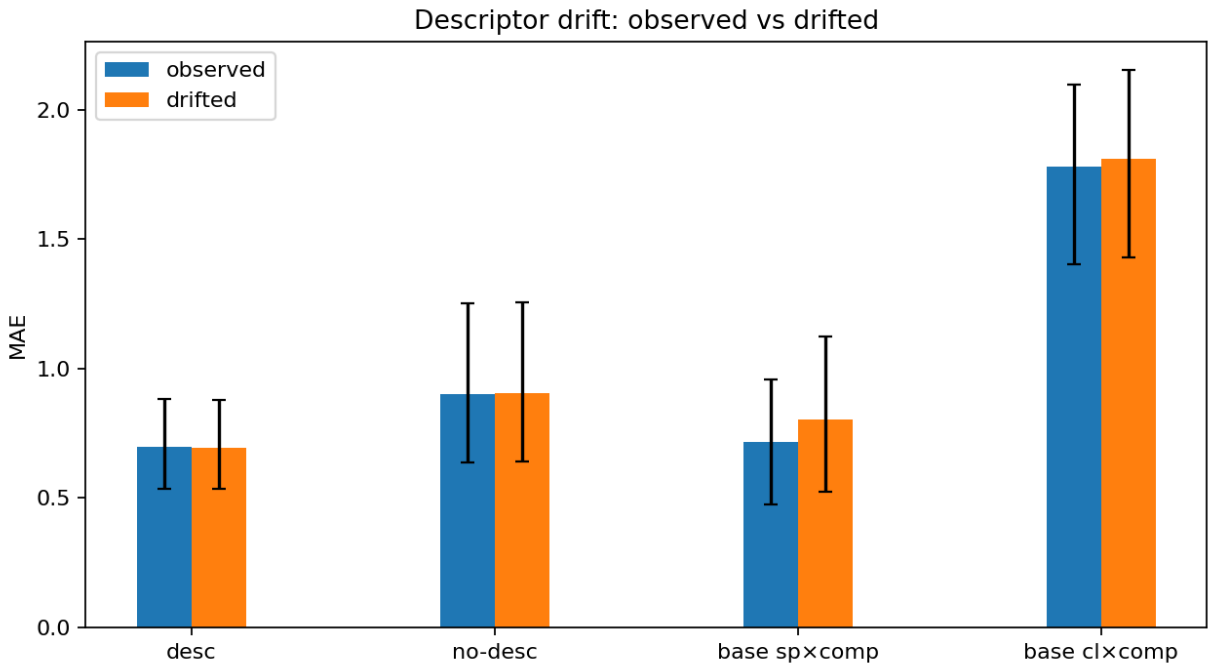
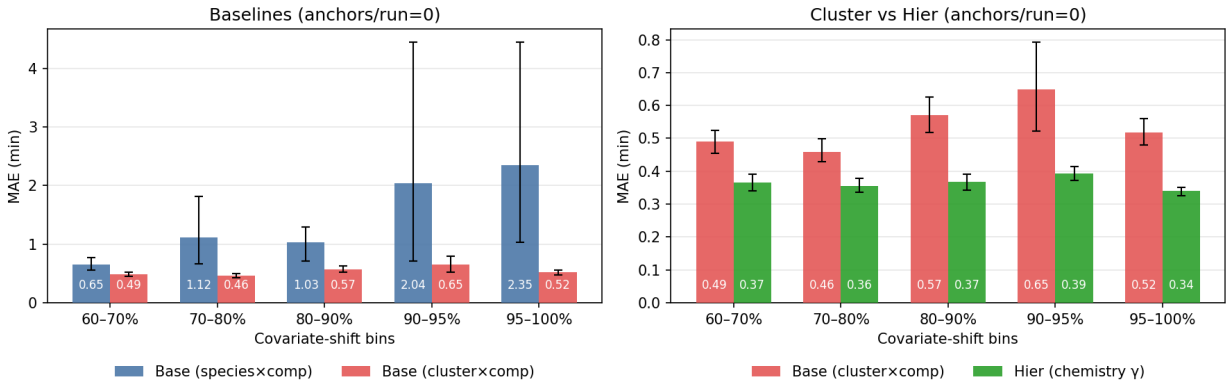


Figure 5: Descriptor-informed hierarchy under synthetic run-level drift. Mean MAE for hierarchical models with and without descriptors, and for species×compound and cluster×compound lasso baselines, evaluated on the same held-out runs with observed covariates and with induced drift. The descriptor-enabled hierarchy degrades more slowly under drift than the chemistry-free hierarchy and lasso baselines.

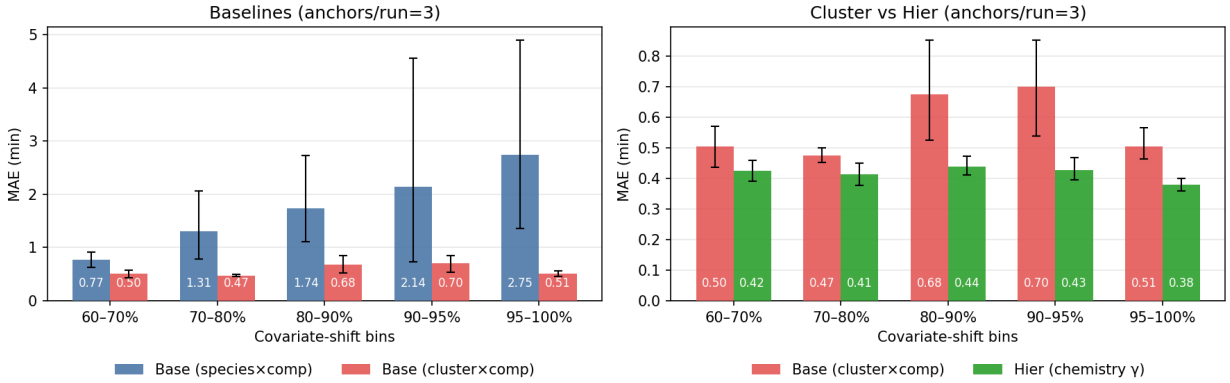
5.2.4 Covariate-shift holdout for historical baselines

Finally, we revisit an earlier covariate-shift benchmark that predates the use of descriptors. In this experiment we generate synthetic RT data with internal-standard covariates, identify in-support runs using a global distance-to-nearest-neighbour score in covariate space, and train the hierarchical RT model (without descriptors) and lasso baselines on those in-support runs. Out-of-support runs are partitioned into distance bins, and all methods are evaluated on each bin.

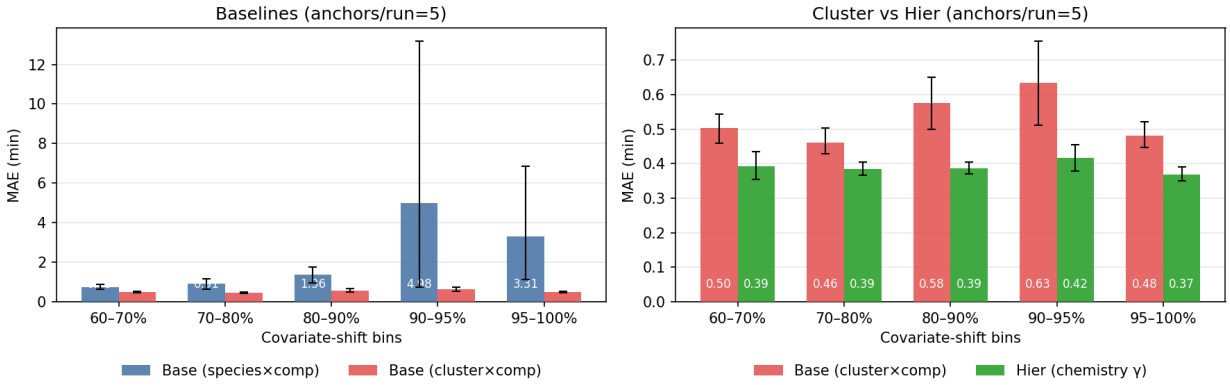
This study provides a baseline view of how the pre-descriptor hierarchy behaves under progressive covariate shift. **Even without descriptors the hierarchical model tends to degrade more gracefully than per-pair lasso fits as runs move further from the training support, especially when a small number of anchors is available to recalibrate intercepts, although all methods incur substantial error in the most extreme out-of-support bins**. The descriptor-focused experiments above build on this foundation to quantify how much additional robustness and generalisation we gain by making the hierarchy explicitly chemistry-aware.



(a) 0 anchors



(b) 3 anchors



(c) 5 anchors

Figure 6: Covariate-shift holdout on synthetic RT. Each panel shows mean MAE (with 95% intervals) for the hierarchical RT model and two lasso baselines across increasing covariate-distance bins, for different numbers of anchor observations per test run. Even without descriptors, the hierarchical model degrades more smoothly under covariate shift than the lasso baselines, and a modest number of anchors substantially improves calibration.

6 Conclusions and future work

We have described a hierarchical RT model for LC–MS that pools information across species, compounds, and runs while retaining interpretable components such as species effects, descriptor-informed compound baselines, and class-pooled run-level covariate sensitivities. By replacing per-(species, compound) lasso fits with a single multi-task hierarchy, the model provides a coherent framework for borrowing strength across related tasks and for obtaining calibrated uncertainties.

On production-style data the hierarchical model (in its descriptor-based configuration with class-pooled covariate effects) matches or improves upon the existing lasso baseline across most species groups while maintaining nearly complete compound coverage. Synthetic experiments show that descriptors are particularly valuable for rare and unseen compounds and for held-out chemistry classes, and that the hierarchy degrades more gracefully than lasso baselines under run-level covariate drift and covariate shift. Together, these results indicate that the hierarchical RT model, with a chemistry prior based on descriptors and classes used only to share run sensitivities, is a promising backbone for future RT modelling in settings with sparse chemistry, diverse species groups, and evolving instrumentation.

At the same time, several important pieces of work remain before this model can be treated as a fully mature production component:

- **Systematic convergence diagnostics for production fits.** The lib208 results presented here are based on NUTS runs that were not exhaustively checked for convergence (e.g. detailed R-hat, ESS, and divergence inspection per group). We need a routine convergence review and alerting pipeline for all production fits, including stress tests on difficult species groups and chemistries.
- **Scalable inference on modern hardware.** Full NUTS on large libraries remains expensive. We should investigate GPU-backed samplers (for example via JAX/NumPyro) and carefully designed approximate methods such as variational inference or structured Laplace approximations. These approaches are non-trivial to deploy safely but could provide the order-of-magnitude speedups needed for routine retraining and cross-validation.
- **Principled propagation of RT uncertainty into peak assignment.** Downstream peak assignment currently relies on relatively ad hoc RT windows tuned for lasso. A key next step is to integrate RT predictive means and variances into the peak-assignment objective so that assignments are scored using the full predictive distribution rather than hard thresholds, and so that uncertainty is carried through to downstream confidence metrics.
- **Cross-instrument and cross-kit generalisation.** The experiments here focus on a single platform (Thermo-trained libraries). For 2026 we need to explore cross-instrument modelling—for example, hierarchical structures with instrument- level effects or shared descriptor priors—and assess how well RT predictions transfer to new instruments such as Agilent and to new kit designs with different chemistry and matrix mixes. Concrete options include instrument-specific offsets on the global intercept and descriptor weights, shared descriptor coefficients with partial pooling across instruments, and using anchor compounds measured on multiple platforms to identify and correct systematic cross-instrument shifts.

- **Model refinements in high-data regimes.** The hierarchical model slightly lags the lasso baseline in very well sampled, homogeneous groups such as human blood, where a separate linear model per species–compound pair is close to optimal. To narrow this gap without sacrificing behaviour in sparse settings, we can introduce a species–compound interaction term (a hierarchical intercept $\delta_{s,c}$ that shrinks toward zero when few observations are available but behaves like a per-pair correction when data are plentiful), experiment with descriptor-informed covariate slopes for run features in place of or in addition to the current class-pooled structure, and tune caps or priors on a per-group basis so that especially data-rich groups can make fuller use of their additional information.

Addressing these items will turn the current RT hierarchy from a promising component into a robust foundation for kit deployment across rare species, diverse chemistries, and new instruments.